Alice 3

Chapter 3 Highlights

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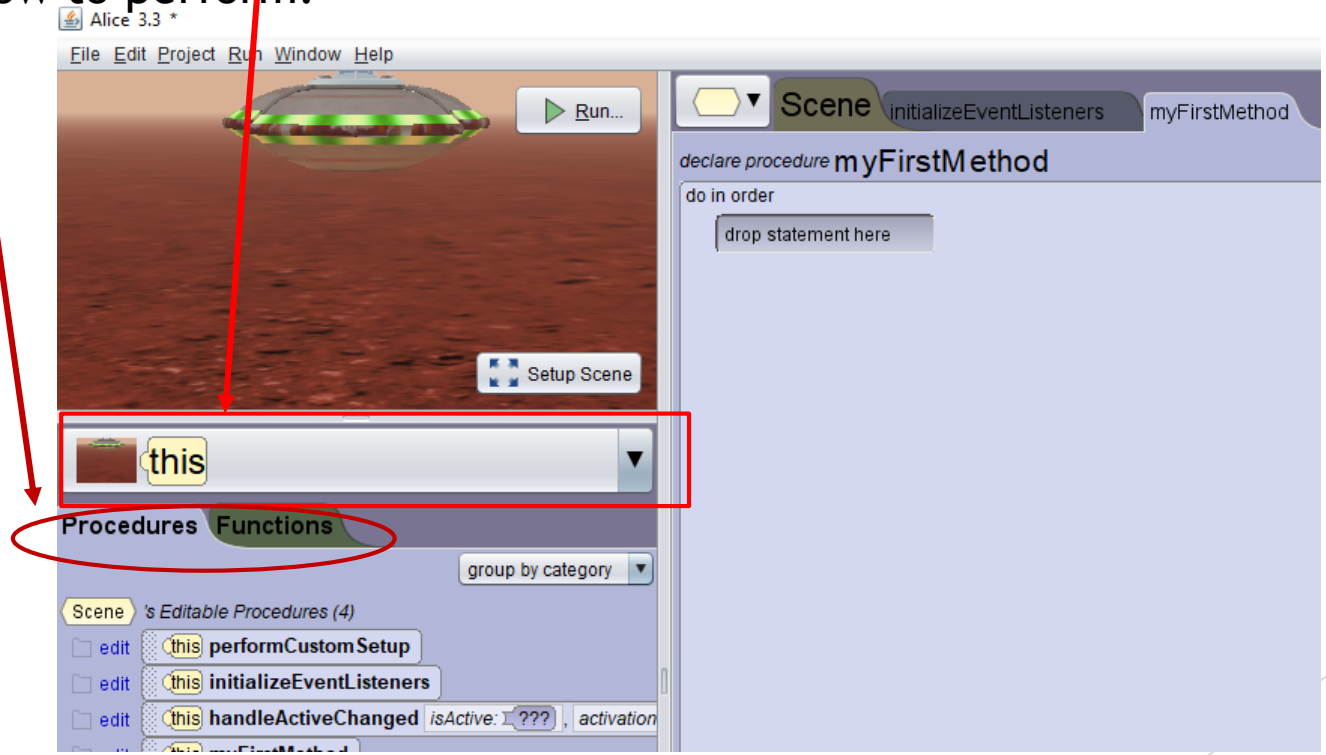
The Code Editor's **Object Selector Box** – Box on the left side of the screen in the code editor that you use to select the Object that you wish to work with.

When you select an object with the object selector box the **Procedures and Functions tabs** contain tiles that represent actions that the selected object "knows" how to perform.



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- A **code statement** is created (written) by dragging a tile into the editor and supplying any information (data) needed to perform the action. When you drag a procedure tile into the editor tab Alice will often ask for some information in a dropdown menu. This is because many procedures have one or more parameters.
- **Parameter** – A piece of information required to perform a task.
- **Argument** – a value sent to a parameter.

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- Scene Class – You can open it by clicking the Scene tab in the right window of the code editor.
- The **Scene class** is a blueprint that tells Alice (1) what actions (procedures and functions) a scene can perform and (2) what components (properties) a scene contains.
- The scene class always has two built in properties (ground and camera). **When you add objects to the Scene they become properties to the Scene**.
- **Active class** – the currently open class.

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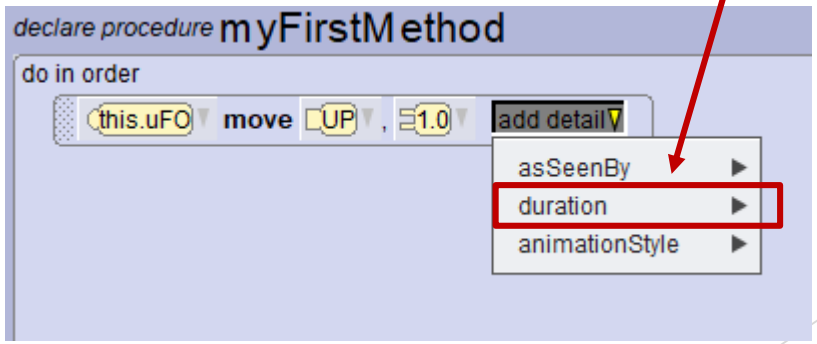
- **Active method** – The code editor tab that displays on top of the other tabs. It is the method currently open for editing.
- **Detail parameter** – Some procedures have parameters that have default values that are called detail parameters. If we do not specify a value for a detail parameter, Alice will use the default value. For example the **duration** is a detailed parameter with a default value of one second. If we do not specify a different duration value (argument) Alice will perform the procedure in one second.



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- **Implementation** – is the process of translating a design algorithm into program code.
- **Keyword** – a word that has a special meaning in a programming language.
- **Method** – Procedures and functions are two different forms of methods.
- **Procedural methods** are actions that can be performed by or on an object.
- **Functional methods** are actions that compute or retrieve information about an object.
- The **Code editor's method panel** (immediately below the Object selector) has two tabs, one containing Procedures (procedural methods) and the other containing Functions (functional methods).

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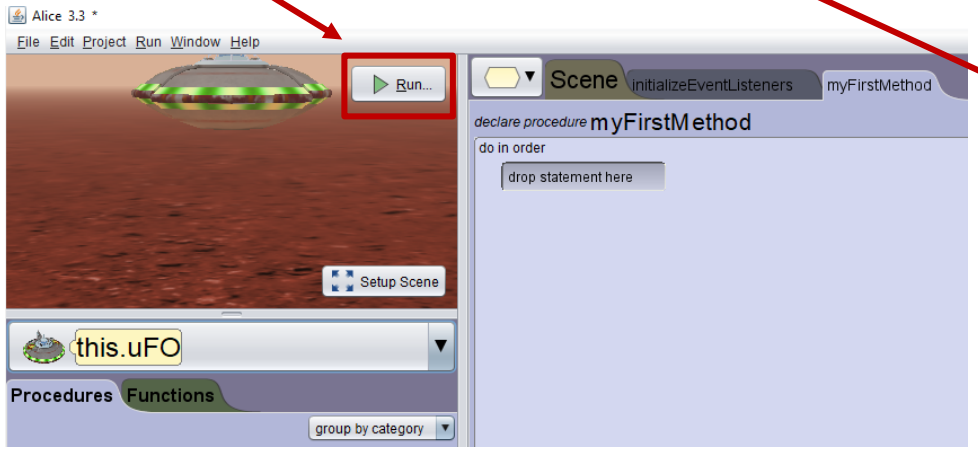
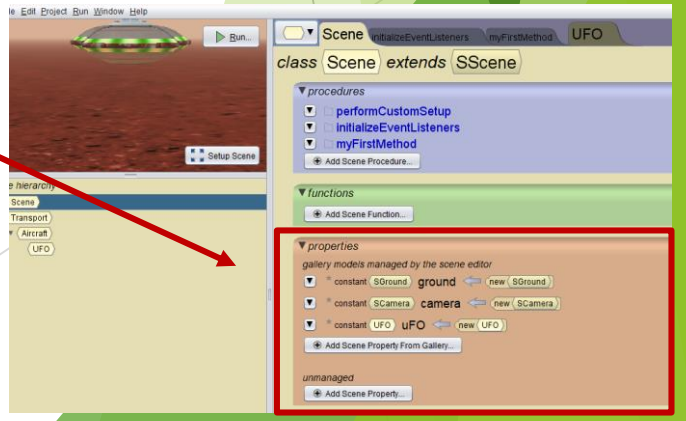
- **Property** – The properties section contains the declaration statements for the objects belonging to a scene.
- **Run button** – Test a program (code) by clicking the Run button. When the Run button is clicked, the program creates a runtime window in which the initial scene is displayed. When the scene is displayed we say the scene has been activated. When the scene is activated, the code statements in myFirstMethod are executed.





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- **Do in order** – the do in order at the beginning of myFirstMethod tells Alice to execute the code statements in sequence, one after the other. The execution of a method (procedure or function) is referred to as calling it.

- To **call a method** means to tell the computer at runtime to perform the action defined by that procedure.

- **Syntax** – In a programming language, syntax is the set of rules that govern the arrangement of words, phrases, and symbols in code statements. The general format (syntax) of an Alice statement is the name of an object followed by the name of a method that is to be performed.